QP code **25900744**

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) REGULAR EXAMINATION NOVEMBER 2025
FIRST SEMESTER
Multi-Disciplinary Courses (MDC) - MG1MDCBCH100 - SPORTS BIOCHEMISTRY: THE SCIENCE OF EXERCISE AND HUMAN PERFORMANCE
(2024 ADMISSION ONWARDS)

Duration: 1 Hour Maximum Marks: 35

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I)and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overalloutcome attainability.

Part A

Multiple Choice Questions
Answer **All** Questions
Each Question carries **1** mark

1. Which of the following is the first biochemical event in tissue repair follows a sports injury?

   Ans- Option a) Clot formation

2. Which of the following functions do nutrients in food serve in the body?
   Ans- options a) and b) (Give marks for attempt)
3. In high-intensity intermittent sports, what energy system is predominantly used?
   Ans- ATP-PC system
4. What is the primary goal of an individualized training program for an athlete?
   Ans -Option c) design a training plan that addresses the athlete's specific needs goals and physical condition.
5. Which of the following is a key enzyme released during muscle damage?
   Ans- Option b) Creatine kinase

[5x1 = 5]

Part B

Fill in the Blanks
Answer **All** Questions
Each Question carries **2** marks

6. Motion capture systems help to visualise the athlete's movement and quantify
   Ans- biomechanical parameters (give marks for attempt)
7. In sports Biochemistry……refers to the study of how genesand physical activity interact to influence athletic performance.
   Ans-Genetics/epigenetics
8. The energy system active for exercise of more than two minutes duration is …….
   Ans.- Oxidative Phosphorylation
9. ……involves visualising successful outcomes to improve motivation and reduce anxiety before events.
   Ans- Visualisation
10. The …….phase of sleep is most important for physical recovery.
    Ans- Deep sleep (slow-wave sleep)

[5x2 = 10]

Part C

Short Answer Questions
Answer any **5** Questions
Each Question carries **3** marks

11. Recall hydration and hydration strategies during exercise.

Ans-**Before exercise :**Precooling Improves endurance performance (cool shower/baths/vests) . 500ml every 2-3hrs prior to exercise (Water, Juice, Milk, Cordial, sports drink) and 5ml/kg immediately before exercise.

During – 250ml every 15-20mts or as much is comfortable during exercise (Water, Sports Drink)

After- 150% of fluid deficit Aim for 500ml to 100ml (sports drink as the electrolytes help stimulate the thirst drive)

(1 mark each)

12. Illustrate the main source for replenishing glycogen after exercise.

- Ans-Carbohydrates replenish glycogen stores depleted during exercise.**(1 mark)**
- **Sources:** Whole grains, fruits, vegetables, recovery drinks.**(1 mark)**
- **Recommended Amount:** 1-1.5 grams of carbohydrates per kilogram of body weight.**(1 mark)**

13. Evaluate the roles of skeletal muscles in the human body.

Skeletal muscles are voluntary muscles attached close to the skeleton. These muscles are attached to the bones with the aid of a fibrous connective tissue called tendon. The primary role of skeletal muscle is for the aid of locomotion and movement. There are mainly two types of skeletal muscles red muscles and white muscle. Red muscles are due to the red pigment called myoglobin, which is in high amounts in the human body. These muscles are smaller in diameter and have a large number of mitochondria in it. The myoglobin stores the oxygen, which is used by the mitochondria for the synthesis of ATP. Red muscles have a large number of blood capillaries in it. Unlike the red muscles, the white muscles are bigger in diameter and have a small amount of myoglobin in it. They also have a smaller number of mitochondria in it. Red fibres are suited for long term slow contraction white fibres are suited for high fibre contraction.

(Function-2 marks Description -1 mark)

14. Why is recovery essential in preventing sports injury.

Preventing Overtraining and Injury

Balanced Training: Avoid overtraining by balancing intense workouts with adequate rest, which helps reduce the risk of injuries.

Monitoring for Signs of Overuse: Be aware of symptoms like persistent pain or unusual fatigue, which may indicate the need for rest to prevent injury.

(1.5 marks each)

15. Design a weekly recovery plan for an athlete based on HRV and sleep tracking data.

Ans-Consistent, high-quality sleep improves athletic performance, reaction times, and decision-making abilities.Aligning training and recovery periods with natural circadian rhythms enhances recovery. For example, optimizing sleep by maintaining a regular sleep schedule helps synchronize the circadian rhythm, leading to better overall recovery.

Heart Rate Variability (HRV)

HRV can provide insights into an athlete's autonomic nervous system balance and recovery status. Monitoring HRV can help determine if an athlete is adequately recovered or overtraining.

Sleep Tracking

Quality sleep is essential for recovery and performance. Using sleep tracking devices or apps can help athletes understand their sleep patterns and identify areas for improvement.

(1.5 mark each)

16. Explain the biochemical effects of thermotherapy.

Ans-Thermotherapy involves applying heat to injured areas to relax muscles, increase blood flow, and promote tissue healing. The Biochemical effects include

Vasodilation: Heat causes blood vessels to dilate, increasing blood flow to the area, which helps deliver oxygen and nutrients to tissues for repair.

Enhanced Metabolism: Increases metabolic activity, promoting the removal of waste products and accelerating the healing process.

Muscle Relaxation: Reduces muscle stiffness and spasms by relaxing muscle fibres and improving flexibility.

(1 mark each)

17. What is the difference between a sprain and a strain.

- Ans-A sprain occurs when ligaments (the tough bands of tissue that connect bones at a joint) are stretched or torn.
- A strain happens when muscles or tendons (the tissues connecting muscles to bones) are overstretched or torn.

(1.5 marks each)

[5x3 = 15]

Part D

Short Essay Type Questions

Answer any **1** Question

Each question carries **5** marks

18. Evaluate the classification of ergogenic aids.

Ans-Ergogenic aids are substances, devices, or practices that enhance an individual's energy use, production, or recovery, thereby improving physical performance.

Nutritional Aids, Pharmacological Aids (Anabolic Steroids, Erythropoietin (EPO),Human Growth Hormone) Physiological Aids (Blood Doping, Oxygen Supplementation), Mechanical Aids(Performance Apparel, Ergonomic Equipment), Psychological Aids (visualization and Hypnosis).

Explain each(1 mark for each)

19. Illustrate the influence of hormones in sports and exercise.

Hormones play a crucial role in regulating energy metabolism, muscle growth, recovery, and overall performance in sports. Key hormones involved in sports performance include:

Insulin:

Function: Regulates blood glucose levels by facilitating glucose uptake into cells, promotes glycogen storage, and inhibits fat breakdown.

Importance: Post-exercise insulin response is critical for glycogen replenishment and muscle recovery.

Glucagon:

Function: Opposes insulin action by promoting

Catecholamines (Adrenaline and Noradrenaline):

Function: Increase heart rate, blood pressure, and energy availability by stimulating glycogenolysis and lipolysis.

Importance: Essential for the fight-or-flight response during high-intensity activities, enhancing performance and focus.

Cortisol:

Function: Increases gluconeogenesis, protein breakdown, and fat mobilization; suppresses the immune system.

Importance: Helps maintain energy supply during prolonged stress or exercise but can impair recovery and immune function if chronically elevated.

Growth Hormone (GH):

Function: Stimulates protein synthesis, muscle growth, and fat metabolism; promotes tissue repair.

Importance: Crucial for muscle development, recovery, and adaptation to training.

Explain each(1 mark for each)

[1x5 = 5]